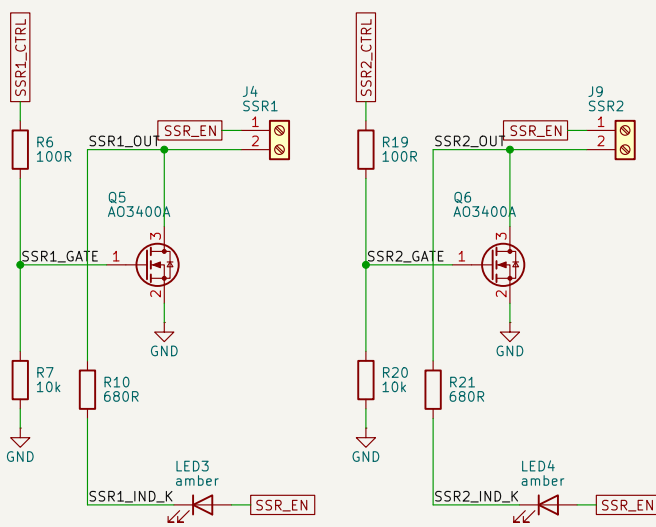
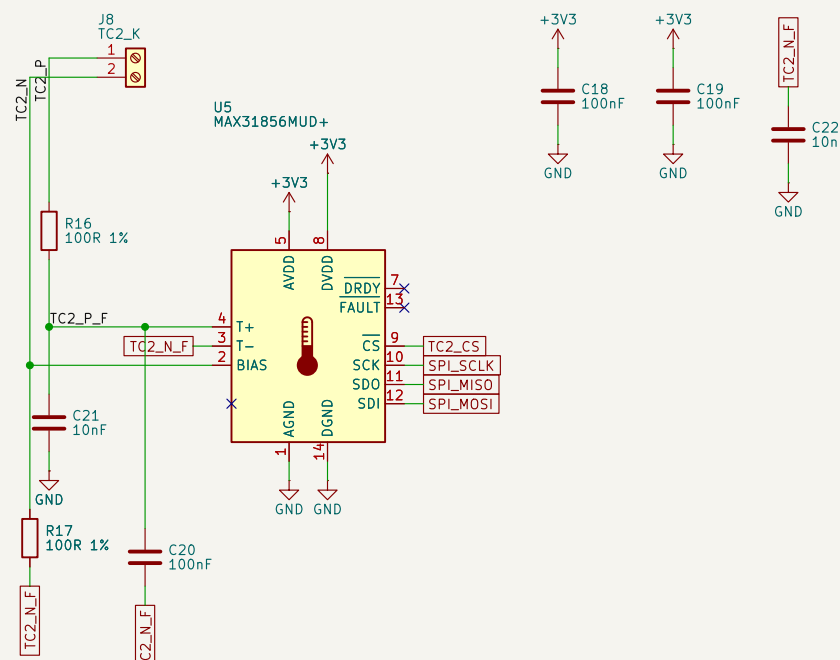
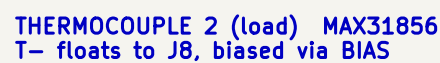
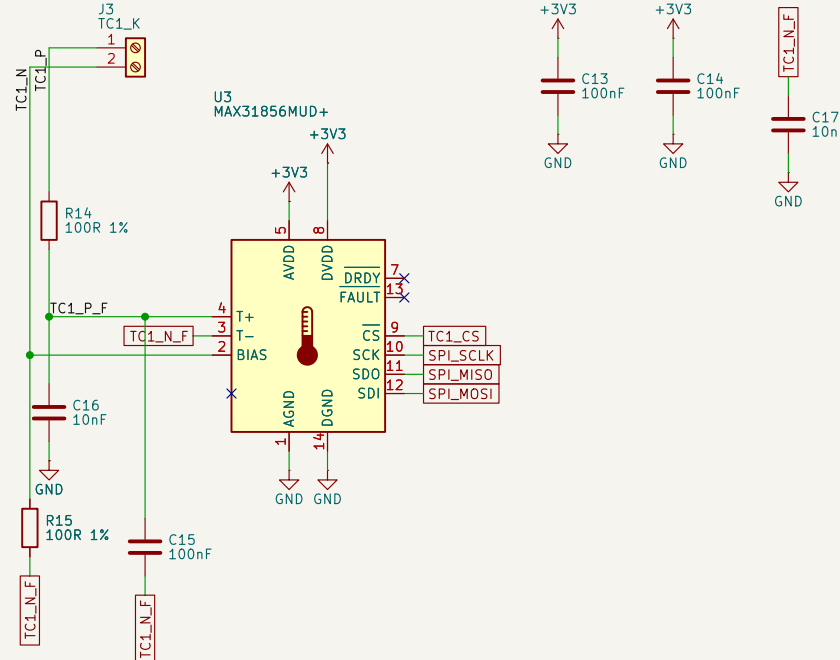
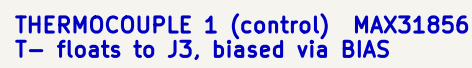
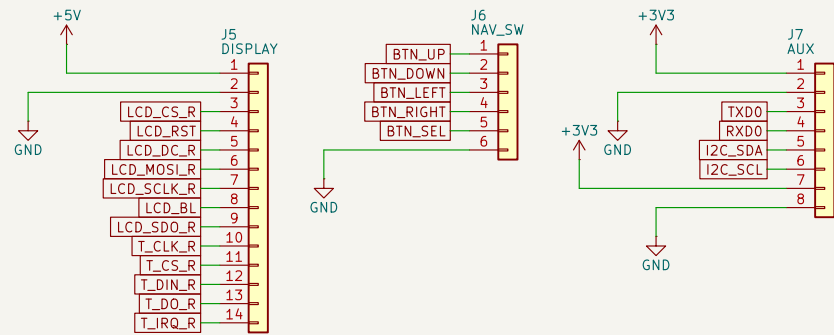
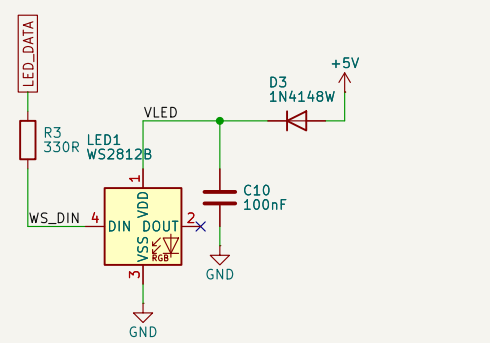
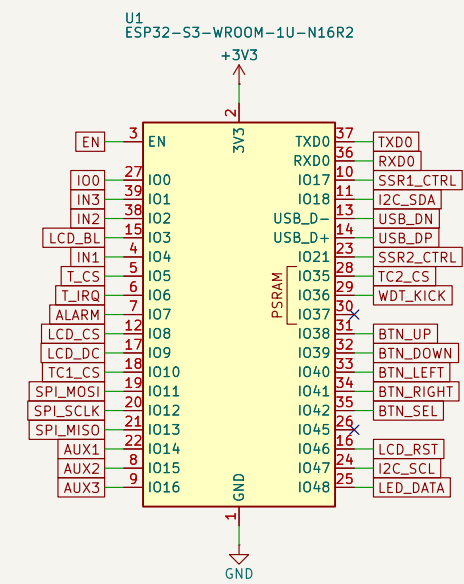
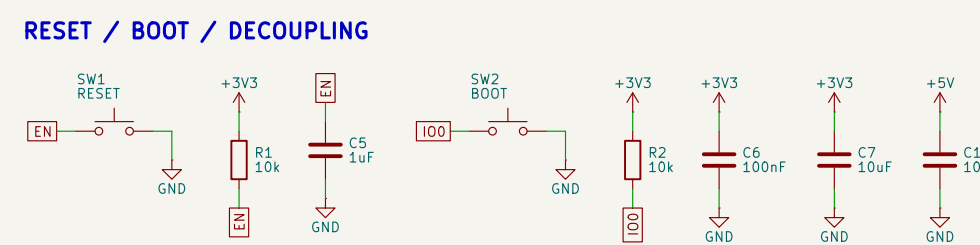
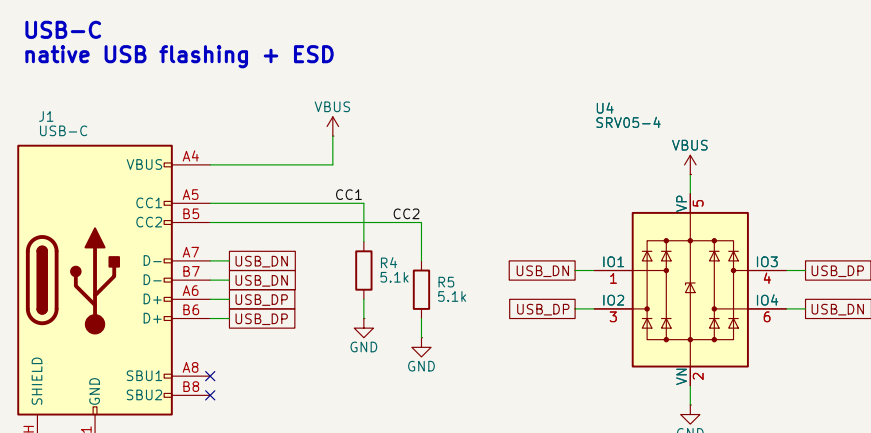
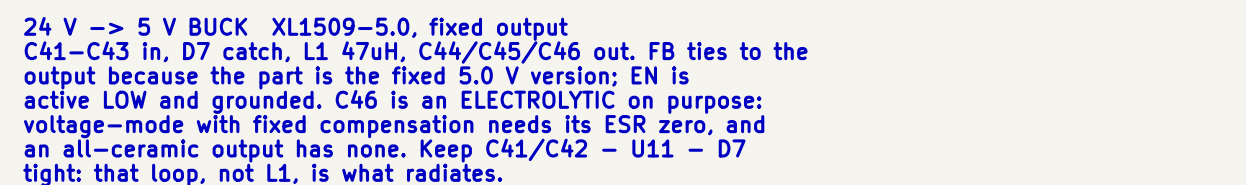
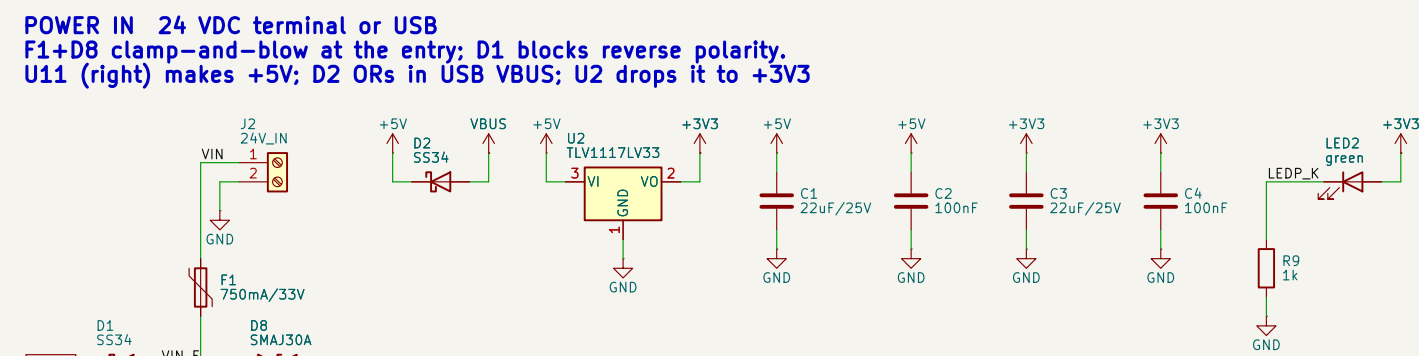
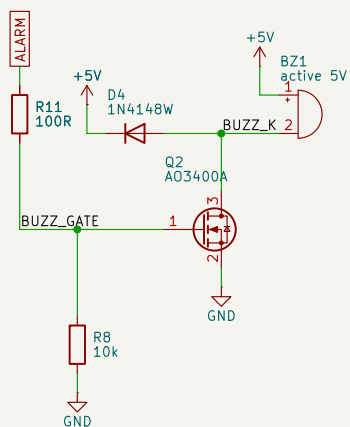
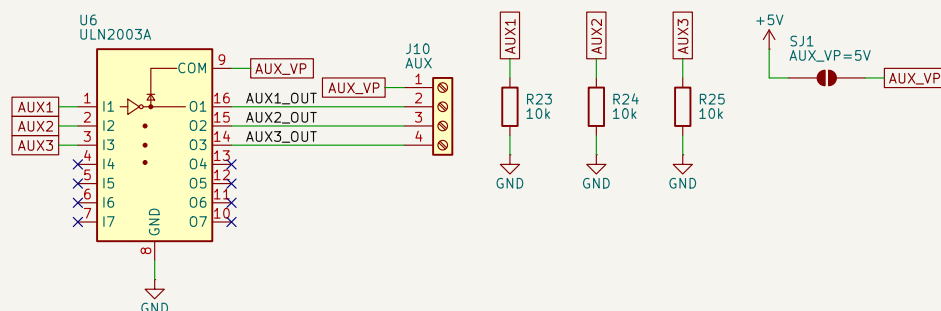
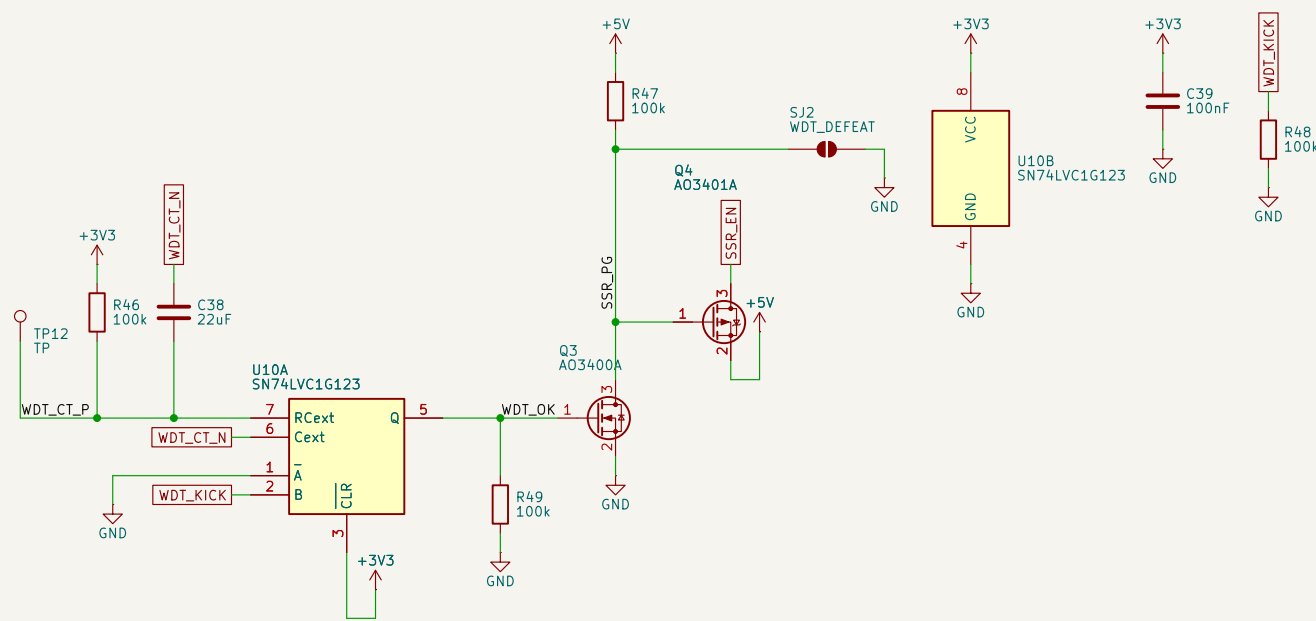


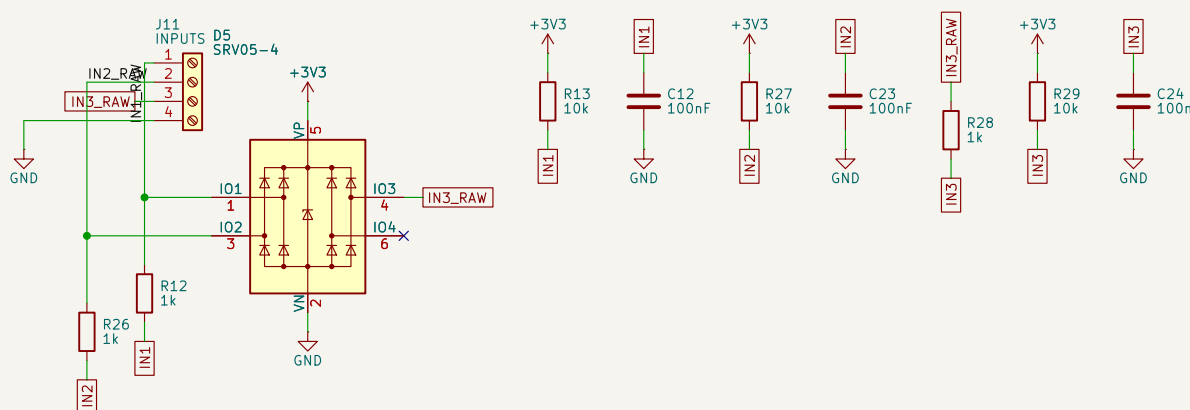


HARDWARE WATCHDOG

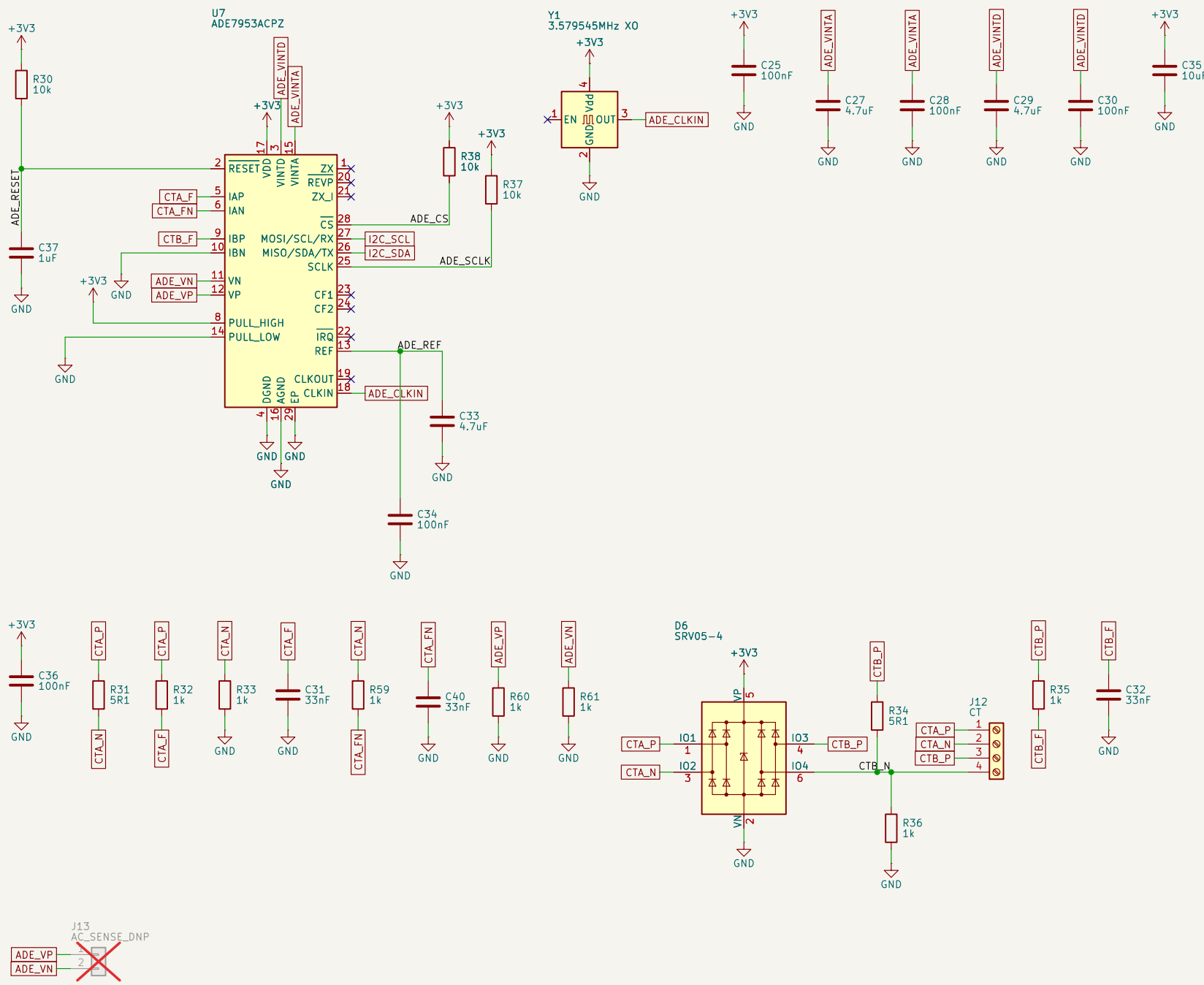
UD10 retriggerable one-shot: every rising edge on WDT_KICK restarts a ~2.2 s window, so Q stays high only while firmware keeps kicking; Q holds Q3 on; Q3 pulls SSR_PG down, turning on high-side Q4, which supplies SSR_EN – the +5V rail feeding BOTH SSR terminals and both indicators. R46/C38 set the window. R48 holds B low when the MCU is high-Z. R47 is the fail-safe pull-up. S/J2 = bring-up defeat, REMOVE for service.



PROTECTED INPUTS x3 lid / gas-flow / spare
1k + 10k pull-up + 100nF each, SRV05-4 TVS

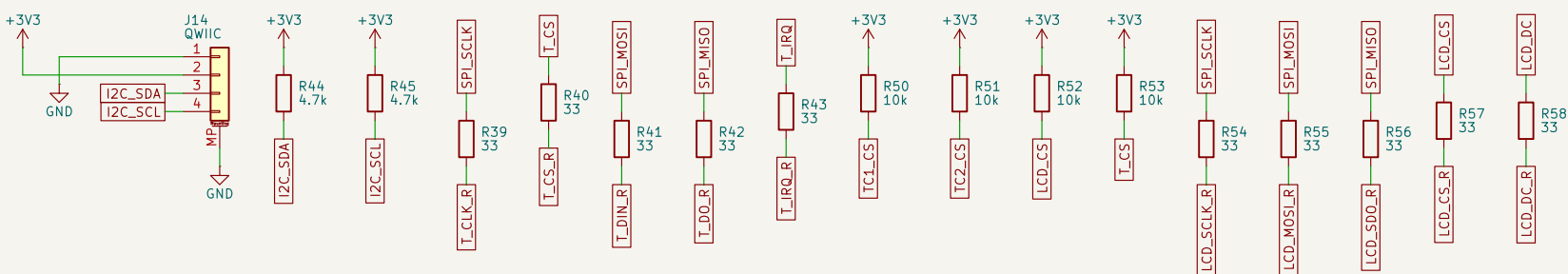


CT CURRENT SENSING ADE7953, I2C, current-only
no mains - VP/VN to DNP J13

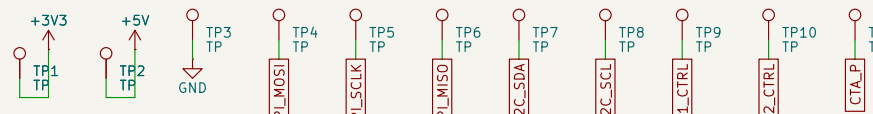


SPI BUS DAMPING + CHIP-SELECT PULL-UPS + I2C EXPANSION

R39-R43 dampie display module's XPT2046 touch lines and R54-R58 the four 40 Mhz display lines; both loom to J5. R56 is also the SDO hedge - lift it if the panel does not tri-state J5.9 off the chip SPLMISO. R59-R53 hold all four chip selects deasserted from power-on, before firmware configures any GPIO. J14 Qwlic and J7 5-8 (0.1 in) share I2C, pulled up by R44/R45.



TEST POINTS



MOUNTING / FIDUCIALS / POWER FLAGS

FID1-3 are bare-copper pick-and-place targets, on no net.



NOTES – Bisque kiln controller, rev E

SSR terminals J4/J9: pin1 = SSR_EN (board +5V, watchdog-gated), pin2 = the switched low side. Boot-safe: R7/R20 hold each MOSFET gate down while the ESP32 pins are high-impedance. NOT opto-isolated - rev B tried that and reverted it: an opto only isolates if the SSR loop is powered off-board, and this board powers it.

TC1 terminal J3 / TC2 terminal J8: pin1 = K+, pin2 = K- - both float, biased near AGND only through each MAX31856's internal BIAS network. Ungrounded-junction probes required: two grounded-junction probes in one kiln would loop through it.

J11: IN1/IN2/IN3 (lid/gas-flow/spare) + GND = dry contact each channel to GND.

Display J5 (14-pin, ST7796S + XPT2046 touch module, LCDWIKI MSP4021): 1=+5V 2=GND 3=CS 4=RT 5=DC 6=SDI/MOSI 7=SCK 8=BL 9=SDO/MISO 10=T_CLK 11=T_CS 12=T_DIN 13=T_DO 14=T_IRQ. PIN 1 IS +5V. NOT 3V3 - the module regulates on board; do not wire a 3.3V-only panel to it. Logic is 3.3V; touch shares SPI2 with the LCD and both MAX31856s (R19-R43 damping).

J12: CT current inputs, CTA_P/CTA_N/CTB_P/CTB_N – one CT clamp per SSR zone.

J13: DNP SELV voltage sense header. NOT mains-rated, not fitted
- do not wire to AC mains. Y1/C25/C26 load caps are an ASSUMED
value (unverified C_L), see design.py comment. ADE7953 I2C
address 0x38 collides with PCF8574A.